

# Research Notes on Autonomous vehicle

## Autonomous vehicle

### >> Section 1

**British definitions** In British English, the word "automated" has several meanings, such as in the sentence: "Thatcham also found that the automated lane keeping systems could only meet two out of the twelve principles required to guarantee safety, going on to say they cannot, therefore, be classed as 'automated driving', preferring 'assisted driving'". The first occurrence of the "automated" word refers to a UNECE automated system, while the second refers to the British legal definition of an automated vehicle. British law interprets the meaning of "automated vehicle" based on the interpretation section related to a vehicle "driving itself" and an insured vehicle. In November 2023 the British Government introduced the Automated Vehicles Bill. It proposed definitions for related terms:

### >> Section 2

Vehicles can potentially benefit from communicating with others to share information about traffic, road obstacles, to receive map and software updates, etc. ISO/TC 22 specifies in-vehicle transport information and control systems, while ISO/TC 204 specifies information, communication and control systems in surface transport. International standards have been developed for ADAS functions, connectivity, human interaction, in-vehicle systems, management/engineering, dynamic map and positioning, privacy and security. Rather than communicating among vehicles, they can communicate with road-based systems to receive similar information.

### >> Section 3

**Self-driving** The Union of Concerned Scientists defined self-driving as "cars or trucks in which human drivers are never required to take control to safely operate the vehicle. Also known as autonomous or 'driverless' cars, they combine sensors and software to control, navigate, and drive the vehicle." The British Automated and Electric Vehicles Act 2018 defines a vehicle as "driving itself" if the vehicle is "not being controlled, and does not need to be monitored, by an individual". Another British government definition stated, "Self-driving vehicles are vehicles that can safely and lawfully drive themselves".

### >> Section 4

**Path planning** Path planning finds a sequence of segments that a vehicle can use to move from origin to destination. Techniques used for path planning include graph-based search and variational-based optimization techniques. Graph-based techniques can make harder decisions such as how to pass another vehicle/obstacle. Variational-based optimization techniques require more stringent restrictions on the vehicle's path to prevent collisions. The large scale

path of the vehicle can be determined by using a voronoi diagram, an occupancy grid mapping, or a driving corridor algorithm. The latter allows the vehicle to locate and drive within open space that is bounded by lanes or barriers.

### >> Section 5

**Navigation** involves the use of maps to define a path between origin and destination. Hybrid navigation is the use of multiple navigation systems. Some systems use basic maps, relying on perception to deal with anomalies. Such a map understands which roads lead to which others, whether a road is a freeway, a highway, are one-way, etc. Other systems require highly detailed maps, including lane maps, obstacles, traffic controls, etc.

### >> Section 6

**Espionage** In 2018 and 2019, former Apple engineers were charged with stealing information related to Apple's self-driving car project. In 2021 the United States Department of Justice (DOJ) accused Chinese security officials of coordinating a hacking campaign to steal information from government entities, including research related to autonomous vehicles. China has prepared "the Provisions on Management of Automotive Data Security (Trial) to protect its own data". Cellular Vehicle-to-Everything technologies are based on 5G wireless networks. As of November 2022, the US Congress was considering the possibility that imported Chinese AC technology could facilitate espionage. Testing of Chinese automated cars in the US has raised concern over which US data are collected by Chinese vehicles to be stored in China and any link with the Chinese communist party.

### >> Section 7

**Approaches** ACs can be tested via digital simulations, in a controlled test environment, and/or on public roads. Road testing typically requires some form of permit or a commitment to adhere to acceptable operating principles. For example, New York requires a test driver to be in the vehicle, prepared to override the ADAS as necessary.

### >> Section 8

**Pedestrian reactions** In 2023 David R. Large, senior research fellow with the Human Factors Research Group at the University of Nottingham, disguised himself as a car seat in a study to test people's reactions to driverless cars. He said, "We wanted to explore how pedestrians would interact with a driverless car and developed this unique methodology to explore their reactions." The study found that, in the absence of someone in the driving seat, pedestrians trust certain visual prompts more than others when deciding whether to cross the road.

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Autonomous vehicle

## >> Section 9

Driving systems Advanced driver-assistance systems (ADAS) automate specific driving features such as Forward Collision Warning (FCW), Automatic Emergency Braking (AEB), Lane Departure Warning (LDW), Lane Keeping Assistance (LKA) or Blind Spot Warning (BSW). An ADAS requires a human driver to handle tasks that the ADAS does not support. ADAS contrasts to an automated driving system (ADS), which would be classified by SAE J3016 as Level 3 or higher.

## >> Section 10

Total incidents The US National Highway Traffic Safety Administration began mandating incident reports from autonomous vehicle companies in June 2021. Some reports cite incidents from as early as August 2019, with current data available through June 17, 2024. There have been a total of 3,979 autonomous vehicle incidents (both ADS and ADAS) reported during this timeframe. 2,146 of those incidents (53.9%) involved Tesla vehicles.

## >> Section 11

Safety model A safety model is software that attempts to formalize rules that ensure that ACs operate safely. IEEE is attempting to forge a standard for safety models as "IEEE P2846: A Formal Model for Safety Considerations in Automated Vehicle Decision Making". In 2022, a research group at National Institute of Informatics (NII, Japan) enhanced Mobileye's Reliable Safety System as "Goal-Aware RSS" to enable RSS rules to deal with complex scenarios via program logic.

## >> Section 12

Development General Motors is developing the "Ultra Cruise" ADAS system, that will be a dramatic improvement over their current "Super Cruise" system. Ultra Cruise will cover "95 percent" of driving scenarios on 2 million miles of roads in the US, according to the company. The system hardware in and around the car includes multiple cameras, short- and long-range radar, and a LiDAR sensor, and will be powered by the Qualcomm Snapdragon Ride Platform. The luxury Cadillac Celestiq electric vehicle will be one of the first vehicles to feature Ultra Cruise.